is indeed an equivalence relation: note that, since S is finite, there is exactly one such component for each ray. If two rays are equivalent – and only then – they can be linked by infinitely many disjoint paths: just choose these inductively, taking as S the union of the vertex sets of the first finitely many paths to find the next. The set of ends of G is denoted by $\Omega(G)$, and we write $G = (V, E, \Omega)$ to express that G has vertex, edge and end sets V, E, Ω.

$\Omega(G)$

For example, let us determine the ends of the 2-way infinite ladder shown in Figure 8.1.3. Every ray in this graph contains vertices arbitrarily far to the left or vertices arbitrarily far to the right, but not both. These two types of rays are clearly equivalence classes, so the ladder has exactly two ends. (In Figure 8.1.3 these are shown as two isolated dots – one on the left, the other on the right.)

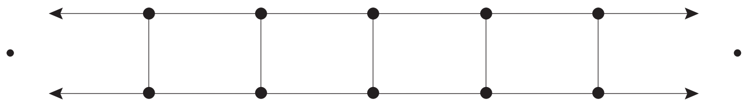

Fig. 8.1.3. The 2-way ladder has two ends

The ends of a tree are particularly simple: two rays in a tree are equivalent if and only if they share a tail, and for every fixed vertex v each end contains exactly one ray starting at v. Even a locally finite tree can have uncountably many ends. The prototype example (see Exercise 39) is the *binary tree* T_2, the rooted tree in which every vertex has exactly two upper neighbours. Often, the vertex set of T_2 is taken to be the set of finite 0–1 sequences (with the empty sequence as the root), as indicated in Figure 8.1.4. The ends of T_2 then correspond bijectively to

binary tree T_2

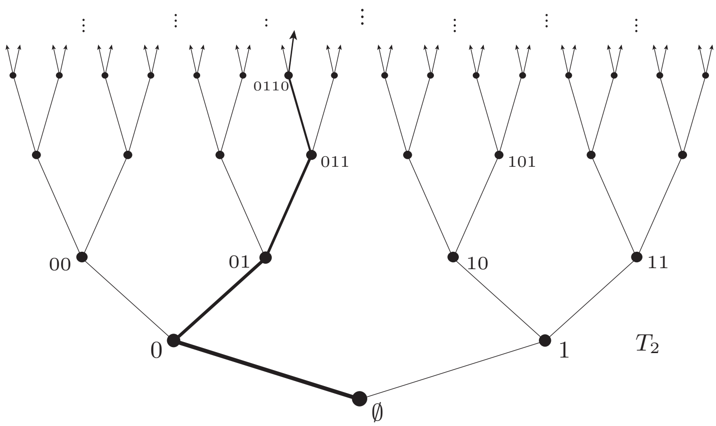

Fig. 8.1.4. The binary tree T_2 has continuum many ends, one for every infinite 0–1 sequence

its rays starting at $\emptyset$, and hence to the infinite 0–1 sequences.

These examples suggest that the ends of a graph can be thought of as 'points at infinity' to which its rays converge. We shall formalize this in Section 8.6, where we define a natural topology on a graph and its ends in which rays will indeed converge to their respective ends.

The maximum number of disjoint rays in an end is the *(combinatorial) vertex-degree* of that end, the maximum number of edge-disjoint rays in it is its *(combinatorial) edge-degree*. These maxima are indeed attained: if an end contains a set of k (edge-) disjoint rays for every integer k, it also contains an infinite set of (edge-) disjoint rays (Exercise 46). Thus, every end has a vertex-degree and an edge-degree in $\mathbb{N} \cup \{\infty\}$. *end degrees*

8.2 Paths, trees, and ends

There are two fundamentally different aspects to the infinity of an infinite connected graph: one of 'length', expressed in the presence of rays, and one of 'width', expressed locally by infinite degrees. The infinity lemma tells us that at least one of these must occur:

Proposition 8.2.1. *Every infinite connected graph has a vertex of infinite degree or contains a ray.*

Proof. Let G be an infinite connected graph with all degrees finite. Let (8.1.2)
v_0 be a vertex, and for every $n \in \mathbb{N}$ let V_n be the set of vertices at distance n from v_0. Induction on n shows that the sets V_n are finite, and hence that $V_{n+1} \neq \emptyset$ (because G is infinite and connected). Furthermore, the neighbour of a vertex $v \in V_{n+1}$ on any shortest v–v_0 path lies in V_n. By Lemma 8.1.2, G contains a ray. $\square$

Often it is useful to have more detailed information on how this ray or vertex of infinite degree lies in G. The following lemma enables us to find it 'close to' any given infinite set of vertices.

Lemma 8.2.2. (Star-Comb Lemma) [8.6.3]
Let U be an infinite set of vertices in a connected graph G. Then G contains either a comb with all teeth in U or a subdivision of an infinite star with all leaves in U.

Proof. As G is connected, it contains a path between two vertices in U. This path is a tree $T \subseteq G$ every edge of which lies on a path in T between two vertices in U. By Zorn's lemma there is a maximal such tree T^*. Since U is infinite and G is connected, T^* is infinite. If T^* has a vertex of infinite degree, it contains the desired subdivided star.